

OOP Using Java (Monsoon Semester)

Course Project

Superstore Management System

IMPORTANT Instructions:

1. It's mandatory that you attend all the deadlines in this project as per the schedule. No request for rescheduling the demo will be entertained. In case of any unavoidable circumstances you have to take email approval from me well in advance.
2. Each group should choose their project latest by 23:59pm on 15.09.2025. Absolutely no changes are allowed after this deadline. Either of the three members in any group should ensure they choose the project in case another group member is unavailable until this deadline.
3. You MUST have a PRIVATE git repository for your project and every group member should very frequently check in their code in this repository.
4. No extensions will ever be provided. Any submission after the deadline will not be evaluated. If you see any ambiguity or inconsistency in a question, please seek a clarification from the teaching staff.

Plagiarism: All submitted deliverables are expected to be the result of your individual effort. You should never misrepresent someone else's work as your own. In case any plagiarism case is detected, it will be dealt as per rules.

Running a Superstore is not an easy task. The Superstore needs a flexible and intelligent automated system that will help them to streamline quantities of various items, minimizes the cost and also satisfy the customers. The project relates to design and development of such an intelligent management system for the Superstore.

The Superstore has multiple warehouses and a chain of stores located in different cities and towns. The architecture of Superstore is illustrated in Figure 1. One or more stores are linked to a warehouse. The linked warehouse ships items to stores on requirement basis as stores has limited space to stock items. The warehouses in turn fulfils the order of stores from their inventoried items or can seek help from another nearby warehouse (i.e. initiates new order to other warehouse). Additionally, a warehouse also keeps track of its inventoried items. When the quantity of stocked item falls below a threshold level (as specified below), it places an order to respective supplier.

A. The operational requirements of the system are:

1. A warehouse provides items of various categories viz. grocery, apparels, electronics, stationary and cosmetics etc. Each category is further divided into subcategories e.g. grocery has the subcategories like beverages, pulses, flours, grains, dry fruits, edible oils etc. The subcategories contain items. The system should provide means to add items in existing subcategories as well as creating new categories and subcategories. Similarly, a store also keeps items based on categories and subcategories. But a store may contain subset of items in linked warehouse.

2. The system should allow to update and delete the inventory of any items as well as that of categories and subcategories in warehouse as well as store. Update includes increasing/decreasing number of unites/items i.e. inventory inwards/outwards, inventory details such as description, cost etc.
3. There is a cost associated with inventory of items in each category. The cost includes fixed cost per quarter D and carrying cost per unit per quarter H . Also, the demand of items in units per quarter K and this information is maintained in inventory details. The system must enable the addition and updation of these quantities at a warehouse and store.
4. The quantities in #3 above enable to compute the number of units superstore should be adding to inventory order. It is called economic order quantity (EOQ) and is aimed at reducing the total costs of inventory management – including factors like ordering costs, holding costs, and shortage costs. The formula to calculate EOQ is,

$$EOQ = \sqrt{\frac{2 * D * K}{H}}$$

The system should automatically compute EOQ for an item.

5. The warehouses also keep a safety stock which is the emergency stock it needs for an item to endure unexpected occurrences - like seasonal acts. It may be zero for some type of items e.g. perishable items. The formula for computing safety stock is

$$\text{Safety Stock} = (\text{maximum daily usage} * \text{maximum lead time in days}) - (\text{average daily usage} * \text{average lead time in days})$$

here, lead time is the time taken since an order is placed till the order is received. The inventory management system should allow to add/update above daily usage of any item. The system automatically computes Safety Stock for each item. Stores do not keep safety stock.

The system calculates the reorder point of an item using following formula,

$$\text{Reorder Point} = (\text{lead time} * \text{average daily usage}) + \text{Safety Stock}$$

here, lead time is the average lead time.

6. When the current inventory level of any item reaches reorder point, the system should generate an alert of ordering EOQ number of items automatically. In case of stores, the reorder point is zero, so stores should order for a item from the linked warehouse when they run out of stock. The system should be able to generate alert for out of stock items for stores.

B. The system has six different types of users:

1. **Super User:** The super user creates new warehouse and stores, link stores to warehouse, create warehouse administrators. Super user can see data of any warehouse and any store.

2. **Warehouse Administrators:** A warehouse administrator is allowed to manage categories and subcategories of items in warehouse. Management here means addition, updation and deletion. The administrator is also allowed to manage all the costs, demand of items, usages, lead time of each item as mentioned in point#3 and 4 in A above and also handles received orders (i.e. whether the items be shipped to store from warehouse or the order will be forwarded to other warehouse). Besides this, the administrator has privilege to view and do whatever other users of the system can do.
3. **Store Administrators:** A store administrator is allowed to manage categories and subcategories of items in the store. Management here means addition, updation and deletion. The administrator is also allowed to manage all the costs, demand of items, usages, lead time of each item as mentioned in point#3 and 4 in A above. Besides this, the administrator has privilege to view and do whatever other users of the system can do. The administrator also places orders to warehouses.
4. **Warehouse Keepers:** There are multiple keepers in the warehouse. Each of them is responsible for a subset of categories and subcategories of items. The keepers can view the data managed by the warehouse administrator but cannot change it. Also, they are allowed to manage inventory inwards and outwards as explained in point #2 in A above.
5. **Storekeepers:** There are multiple storekeepers in the store. Each of them is responsible for a subset of categories and subcategories of items. The storekeepers can view the data managed by the store administrator but cannot change it. Also, they are allowed to manage inventory inwards and outwards as explained in point #2 in A above.
6. **End Users:** The end users of the Superstore system are actually buyers. They can browse through various categories and subcategories at a store and check the current availability of items. The end users should be provided an option to search any item, sort items alphabetically. The search should support partial matching of search criteria with item names. End users are not allowed to carry out any modifications into the Superstore system.

C. Some other Salient points are:

1. All the users can also access the system remotely.
2. Each warehouse and store are identified by unique id and also have a name.
3. All administrators and keepers have their respective login id and passwords. Only respective administrator can create new users i.e. keepers by creating new login id and passwords for them. The keepers can change their passwords. The login id and password for administrators is created by the super user. The login id and password for super user is created by the application or the system.
4. The end users login to the system as a guest users.
5. Whenever an administrator (warehouse or store) logs in, the system automatically shows alert message to him/her for items that needs to be ordered i.e. the system automatically computes reorder point (#6 above in A) of each item and matches it with current inventory level, if it is same it implies that a new order of level EOQ (point

- #4 in A above) of the item is to be placed. The alert message should be shown in a new pop window and items there in should be sorted based on their reorder points (i.e. items with 0 reorder point first and so on).
6. The items that needs to be reordered are in a list based on priority of reordering at store as well as at warehouse.
 7. A warehouse administrator can see the data of any linked store but store administrator cannot check the data of linked warehouse.
 8. A warehouse administrator can check data of any other warehouse.
 9. Warehouse keepers can check data of only and only their respective warehouses. Similarly, a storekeeper can check data of their respective stores.
 10. The store administrators should able to send a message to linked warehouse for required items. The message should contain a list of records where a record is for one item and contains item name, item code (unique code), quantity, expected arrival date.
 11. The warehouse administrator should able to receive messages from linked stores. All such messages should appear in a window when respective administrator logs into the system.
 12. The warehouse administrator should be able to forward a message received from store administrator to other nearby warehouse if the item is not available in his warehouse.

D. System requirements

The Superstore Management System provides

- (i) Different GUIs for different types of users i.e. the main window of Superstore management system should give any option for use type beside others and then show second window based on their type. Also, different views should display data to users based on their access privileges.
- (ii) Any user can access the system remotely by downloading a client i.e. the system should support client server architecture as illustrated in Figure 2.
- (iii) The files are to be used to store the data. Any exception in the application is logged into a log file. The overall system architecture is shown in Figure 2.
- (iv) An end user should be allowed to select the store and check data of any store. Data in the context of end users mean category, subcategory and name of items.
- (v) Whenever, any user (administrators, keepers or end users) selects a category, the subcategories should appear in new window. Similarly, if they select a subcategory, the items should appear in a new window. An option to sort items should be available in all these windows.
- (vi) The application should allow to open multiple client windows simultaneously i.e. super user, warehouse administrator and/or store administrator window together on one machine.
- (vii) Use Junit for unit testing. All the classes should have corresponding test classes.

Figure 1: Superstore Architecture

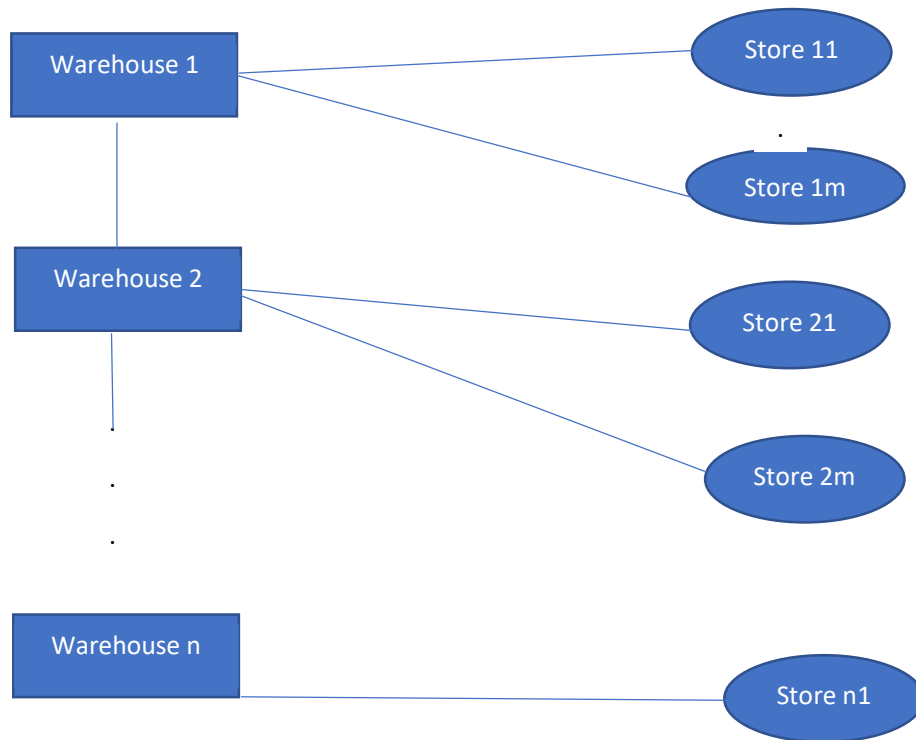


Figure 2: Automated S/w Architecture

